Recommendation Engine for E-Commerce

Bias-Aware Personalization & Dynamic Retargeting

Infosys Springboard Internship — By Sudhansu Kumar Dash

Problem Statement

- Engagement vs fairness

  Recommendation systems drive engagement and conversions — but often favor popular items, reducing diversity.

- Popularity dominance

  Popular products accumulate visibility, creating feedback loops that suppress long-tail discovery.

- Cold start

  New users and items lack personalization signals.

- Recommendation fatigue

  Repeated suggestions reduce satisfaction and limit discovery.

Project Objective

End-to-end system

Build end-to-end recommendation system.

Popularity bias

Introduce methods that reduce popularity bias.

Improve personalization

Give product recommendations to the users based on their interests.

Dynamic retargeting

Threshold-based retargeting logic to prioritize recommendations.

Interactive demo

Deploy an interactive demo where the model will work.

System Architecture

1

Data Layer

Users, Products, Interactions dataset preparation.

2

Processing Layer

Cleaning, scoring, item-user matrix, normalization (L2).

3

Model Layer

Item–Item similarity, KNN-based recommenders, retargeting rules, and bias mitigation modules.

4

Application Layer

Serialized models, Streamlit interface for demo, and live session tracking for personalization.

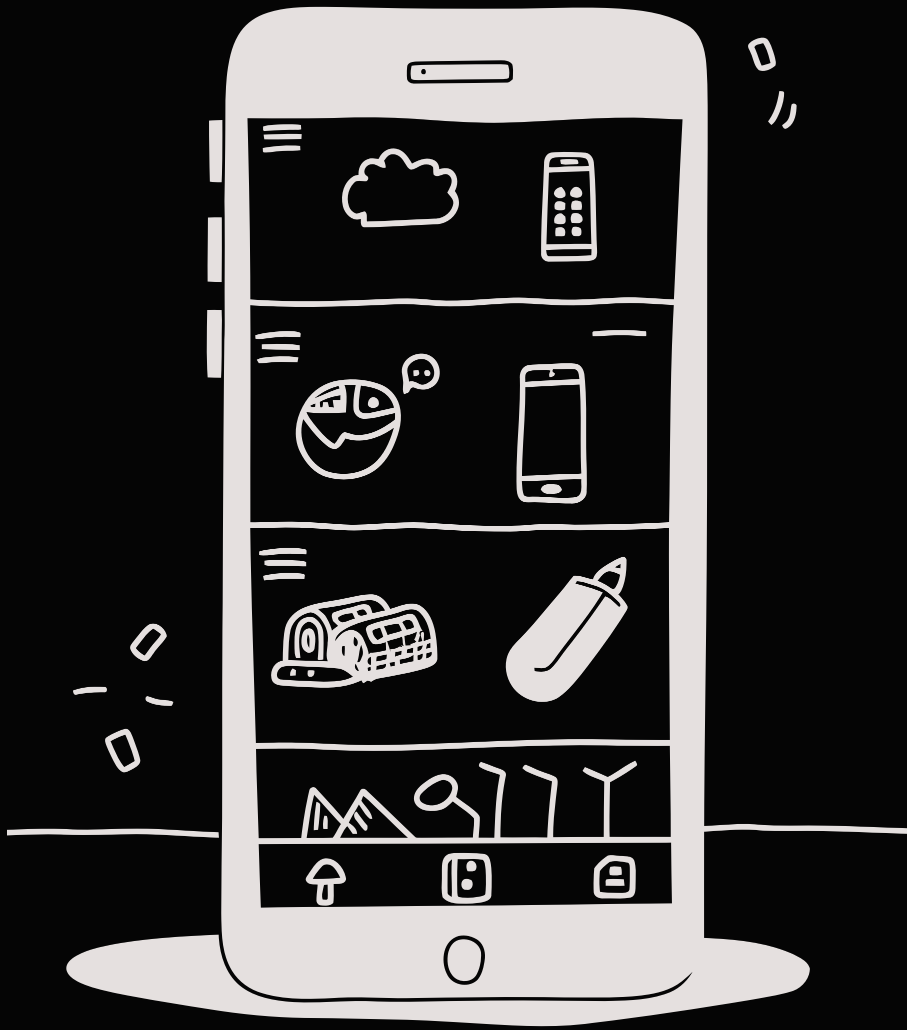
Technology Stack

Data & Processing

- Python
- Pandas
- NumPy

Modeling & Deployment

- sklearn for KNN, normalize L2 and baseline models
- normalize L2 for reducing dominance of popular products.
- Streamlit for interactive demo; Pickle for model serialization



Data Creation — Controlled, Reproducible Datasets

Purpose: Create a controlled and realistic dataset for experimentation.

Deterministic generation

Generation pipelines seeded for reproducibility across runs and environments.

Realistic user profiles

Unique emails and safe test credentials for each user.

Structured products

Programmatically generated products with structured categories, variant naming (e.g., Pro, 2025 edition) and description templates for each.

Price mapping

Base pricing with modifier-based variation for each product to simulate realistic prices.

Data Cleaning & Normalization

Purpose: Prepare noisy data for reliable modeling and downstream analytics.

Naming inconsistencies

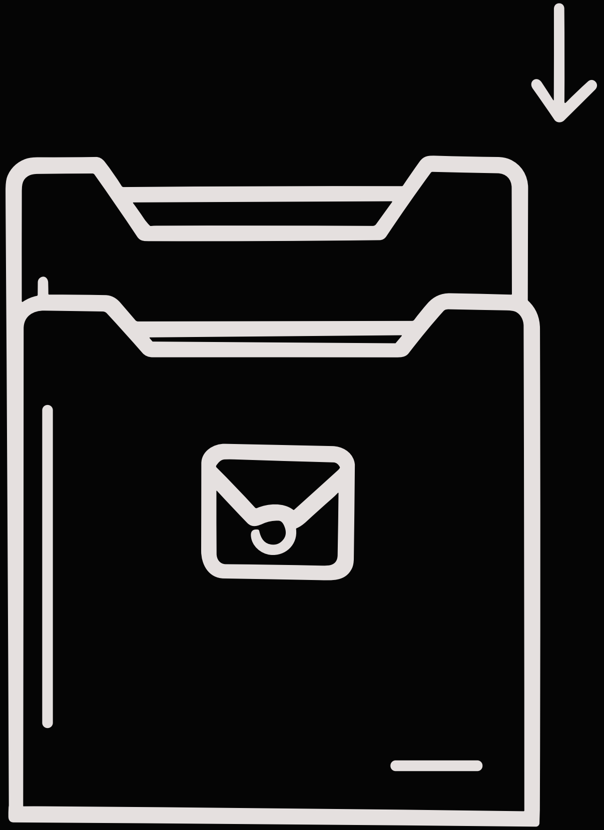
Introduced realistic typos, abbreviations, and casing differences to validate normalization.

Category standardization

Standardized categories using similarity matching

Unique IDs & formatting

Ensured unique user and product IDs and consistent formatting across all fields.



Asset Automation

Purpose: Remove external dependencies and ensure stable runtime for experiments and demos.

Read generated files

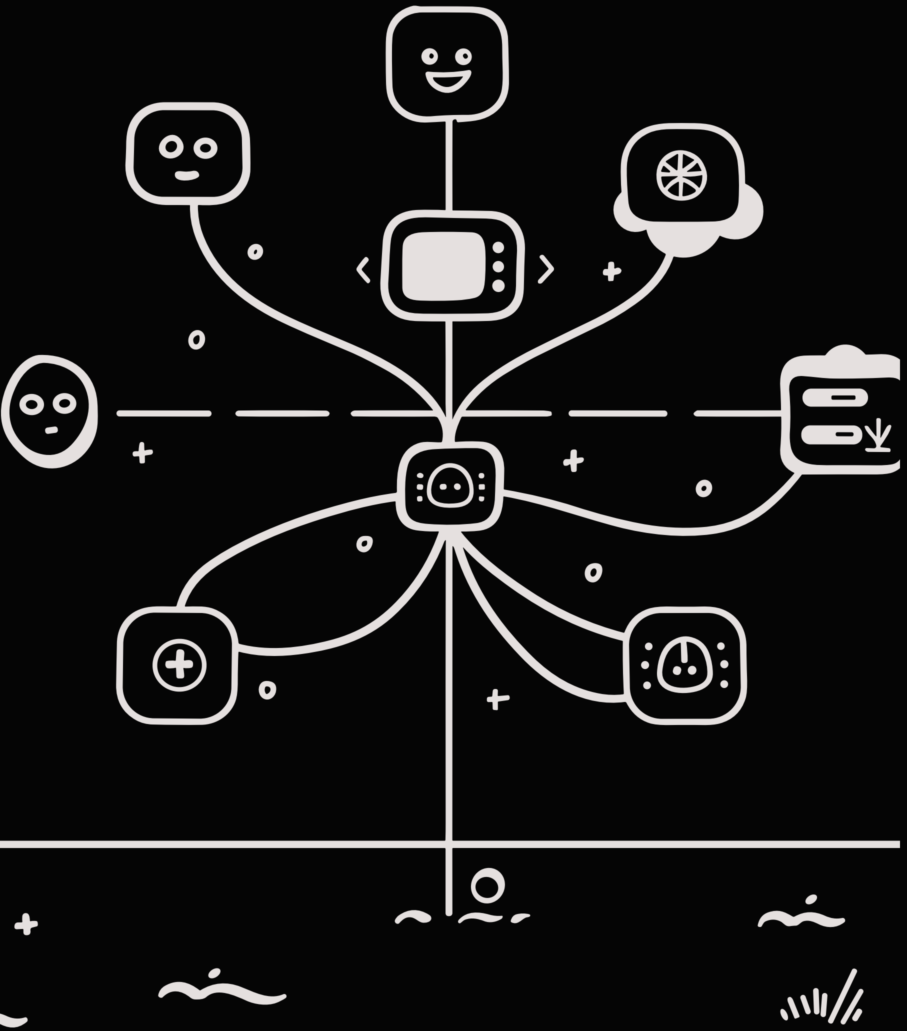
Read product image URLs from product dataset.

Automated retrieval

Automated batch image downloads.

Localization of links

Rewrote external URLs to internal paths and updated the product files.



Interaction Simulation — Persona-driven Behavior

Purpose: Replicate realistic user behavior and controlled skews for model testing.

Persona modeling

Defined Student, Employed, and Non-employed personas for every user.

Event simulation

Simulated view, cart, and purchase events with values for each.

Popularity bias

Injected controlled popularity bias to replicate real-world datasets with overlap.

Interaction Scoring & Matrix Construction

Purpose: Build a stable numeric foundation for collaborative filtering and ranking algorithms.

Action weights

Mapped actions to numeric scores (View=0.5, Cart=3, Purchase=30) to reflect intent strength.

Score controls

Applied score caps to prevent dominance by repeated views.

Interaction matrix

Built item-user matrix from the interactions file to be used by model for recommendations.

Feature Preparation — Ratio based Reweighting

Purpose: Improve novelty and reduce dominance of high-frequency items in recommendations.

sklearn normalize

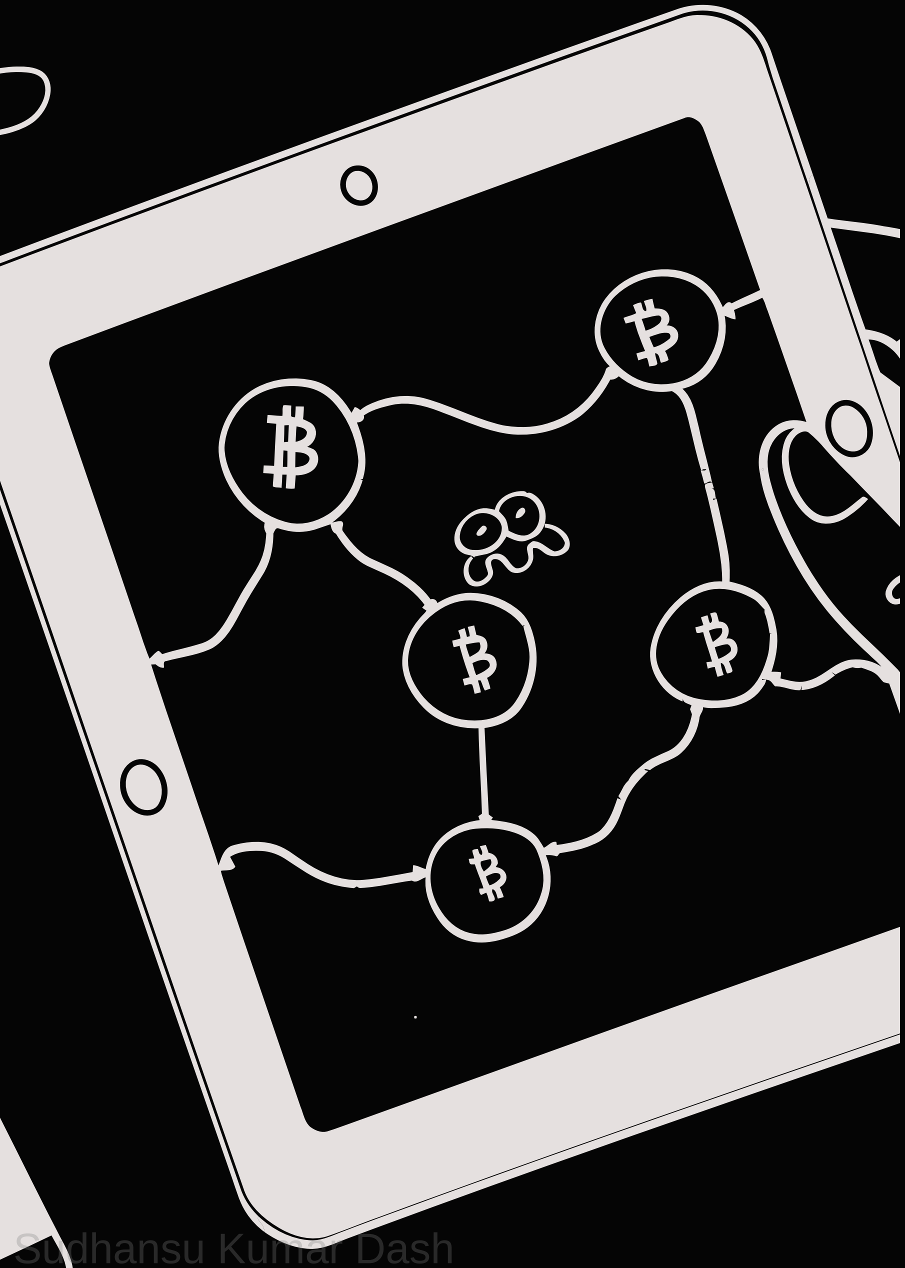
Reduced dominance of popular products, appearing in recommendations of every user.

Normalization

Divided the vector length by the individual scores, balancing out the total interaction scores.

Exposure for long-tail

It boosted lower-frequency items to increase discovery while preserving relevance signals.



Recommendation Model — Efficient Item–Item Approach

Purpose: Deliver relevant Top-N recommendations with predictable latency and explainability.

01

Similarity computation

Cosine similarity on items to find nearest neighbors.

02

Candidate generation

Aggregate the scores to find out the Top N neighbors for a product.

03

Ranking

The top N neigbours are shown in decreasing order of their similarity scores.

Cold Start & Retargeting Strategies

Purpose: Maintain engagement for new users and re-engage users without repeating purchases.

Cold start fallback

Use globally top-interacted items for users without history.

Retargeting

Re-surface items below purchase threshold or previously viewed but not bought.

Exclusions

Filter out purchased, blacklisted items to avoid irrelevant suggestions.

Popularity Bias Mitigation

Purpose: Balance personalization with product diversity to improve downstream KPIs.

Detection

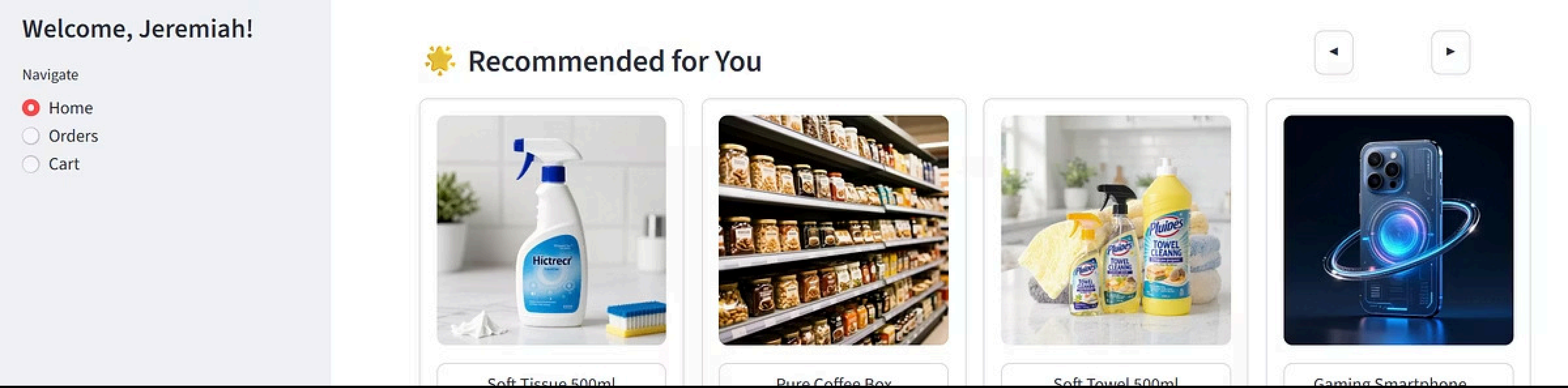
Identify dominant products using the total interactions scores.

Conversion to ratios

Divide the length by individual scores which gives a balanced ratio.

Long-tail lift

Increase long-tail exposure taking the balanced ratios.



App Integration

Purpose: Demonstrate complete end-to-end flow from model artifacts to interactive UI and live tracking.

Serialized artifacts

Serialized model artifacts for faster loading.

Streamlit interface

Multi-page dashboard for exploring recommendations, product categories, product details, cart, orders.

Live session tracking

Live session interaction ranking where each click influence and refresh the recommendations.

Dynamic updates

Dynamic recommendations without any reload.

Key System Features

Persona-driven simulation

Give curated recommendations for each user.

Popularity bias mitigation

Remove popular products appearing in every users recommendations.

Threshold retargeting

Re-surface high-interest items that were viewed or added to cart but not purchased.

Cold-start strategy

Recommendations available for users without any history in datasets.

Real-time personalization

Real-time session personalization with live scoring in each session, dynamically updating recommendations without need of reload.

Deployment & Live Demo

Optimized loading

Serialized model artifacts for fast application startup and responsive user experience.

Streamlined Interface

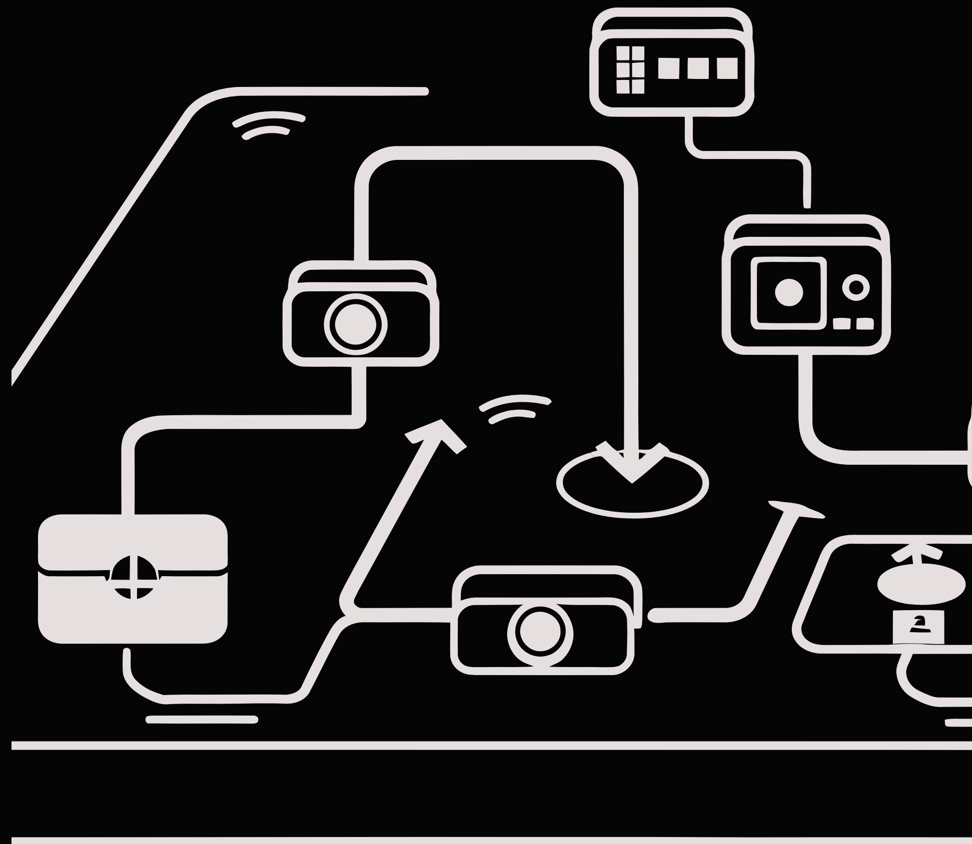
Integrated into a multi-page Streamlit dashboard for intuitive exploration.

Real-time tracking

Live session interaction tracking to understand user behavior and impact.

Dynamic updates

Recommendations adapt instantly based on user actions, providing personalized experience.



Future Scope & Improvements

Database Integration

Move beyond session-based data to capture long-term user behavior.

Integrate real-world datasets

Use publicly available e-commerce data for robust evaluation.

Explore advanced models

Investigate SVD and deep learning-based recommenders for enhanced accuracy.

Develop production frontend

Build a React-based web application for a polished user experience.

Conclusion

Built complete end-to-end recommendation pipeline

Balanced personalization and product diversity

Integrated model into live interactive demo

Thank you — Questions?

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